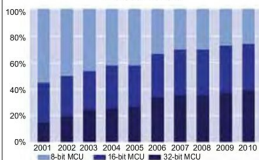


Relative Market Share of 8-, 16-, 32-bit Microcontrollers



Source: iSuppli

demand because of an increasing focus on energy efficiency. The profusion of battery-powered applications is further driving the sales of low-power microcontrollers. Low power is no longer associated with low performance, thanks to the current crop of powerful processors in this segment.

Robustness. Robustness, with electromagnetic interference and electromagnetic compatibility (EMI/EMC) shielding, is another enhancement seen in MCUs. "Microcontrollers need to sustain harsh temperature conditions and have a long lifecycle so that they can be used in both automotive and high-performance industrial applications," shares Vivek Tyagi, country sales manager, Freescale.

According to Duvendage, the saying "the more the things change, the more the things stay the same" is also applicable in the microcontroller market, with customers still demanding 5V operation for robust operation in noisy environments, even though many electronic devices are moving toward lower voltages.

Strong programming support. Tools and software ecosystem is another important differentiating factor for the MCUs. Thapliyal shares, "Today, most of the application-specific software are available in near-product-ready form. It is much more than conventional free software peripheral libraries, flash programming utilities, etc. It's not uncommon now for customers to choose an MCU platform from so many alike-MCU-features vendors because a suite

of reference software is available for that MCU."

"On-chip debug and in-system programming, which were earlier considered as an 'expensive extra,' are feature requirements for most of the users today," shares Eieland. The industry has seen a strong progress for the development tools and tool-chains in past few years. "Most MCU vendors today value the engineers' time and

do their utmost to ensure that the tools are easy to set-up and that engineers can start development fast and finish projects within their deadlines, making them successful in their jobs," Eieland adds. Being able to reuse code across multiple projects is an important part of this.

Human-machine interfaces (HMI).

Today, alphanumeric liquid-crystal displays (LCD) are being replaced with transistor LCDs as the latter offer a far superior user experience. Technological developments have opened completely new user interface options that are capable of a higher level of interaction and engagement with the user. As the display is a highly visible element of any electronic system, such a high-quality HMI offers many advantages over competing products using old-fashioned alphanumeric LCDs.

MCU vendors are commissioning themselves to provide ready-to-use platforms for easy and fast development of HMIs. Flash-based MCUs provide almost zero boot time, making them more preferable for HMI solutions.

"As consumer electronics become more standards-driven, companies are looking for ways to differentiate their products, resulting in several emerging markets for various user interface solutions such as touch sensing and graphics," says Duvendage.

Vendors are working on providing innovative resistive-touch, capacitive-touch and multi-touch single-chip solutions.

Vendor profiles

NXP Semiconductors

NXP Semiconductors offers 8-, 16- and 32-bit MCUs and is focusing on 32-bit ARM architecture with five MCU cores to serve the full range of application requirements. Its leading ARM9/ARM7-based LPC3000 and LPC2000 families have numerous, sophisticated integrated peripherals available. The company also offers low-power, cost-effective ARM Cortex-M3 and Cortex-M0 cores as part of its LPC1000 family, running at speeds of up to 100 MHz and offering best-in-class performance and peripheral support.

ARM-based L77A and L77B families feature high-resolution integrated LCD controllers and provide system-on-chip capability, while LPC900, LPC700 and 80C51 microcontroller families deliver 80C51-based performance. NXP's ARM Cortex-M3 microcontroller (LPC1800) is optimised for low-power operation at very low frequencies all the way through to 150MHz maximum performance from either Flash or RAM.

Renesas Electronics

Renesas' 8- and 16-bit MCUs, such as the R8C, M16C and H8S families, have been very well received in the market for their performance and electromagnetic compatibility. There has been a growing interest toward higher-MIP 32-bit MCUs, such as H8SX, R32C and SH2(A). Renesas has also created a new 32-bit core, named as Renesas eXtreme (RX), after a lot of market application analysis in terms of opcodes, architectures and power consumption to meet the market needs. This CPU core can achieve 200 MHz, 1.65 DMIPS/MHz and 0.03 mW/MHz.

Renesas Electronics standardises all of the peripheral IPs in its MCUs to make them compatible to each of its CPU cores including the 78K, R8C, RX, SuperH and V850.

Motion sensing is another important feature of newer applications and products. The wide range of ultra-low-power MCUs enables designers to easily differentiate portable devices such as smart phones, sports equipment, gaming equipment and bluetooth headsets with motion-sensing technology for gesture recognition, image stabilisation, etc.

Multi-core. "Multi-core MCUs